Chemistry - Graduate Level Assessment 4 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Evaluate the series $S = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$ $

   (a) 0.5
   (b) 2.5
   (c) 1.25
   (d) 1.5
   (e) 4

2. A chemist wants to produce chlorine from molten NaCl. How many grams could be produced if he uses a current of one amp for 5.00 minutes?

   (a) 0.031 g
   (b) 5.55 g
   (c) 0.021 g
   (d) 0.511 g
   (e) 0.111 g

3. 2.000 picogram(pg) of $^33Pdecaysby^0\_{-}1\beta emissionto0.250pgin75.9days.Findthehalf-lifeof^33P.$

   4. 37.95 days

   5. 50.6 days

   6. 8 days

   7. 5.6 days

   8. 60.4 days

The photoelectron spectrum of carbon has three equally sized peaks. What peak is at the lowest energy?

(a) The 2 p peak has the lowest energy.
(b) The 3 d peak has the lowest energy.
(c) The 2 d peak has the lowest energy.
(d) The 1 d peak has the lowest energy.
(e) The 1 p peak has the lowest energy.

Calculate the voltage of the cell Fe; Fe^{+}2$||H^+; H_2iftheironhalfcellisatstandardconditionsbuttheH^+ionc$

0.50 volts

0.76 volts

1.2 volts

0.33 volts

.85 volt

True / False (10 marks)

Answer True or False

6. True or False: Thionyl chloride (SOCl 2) has a tetrahedral molecular geometry due to the arrangement of its four electron pairs.

7. True or False: The Q-factor for an X-band EPR cavity with a resonator bandwidth of 1.58 MHz is 9012.

8. True or False: The value of $(0.0081)^{\{}3/4\}isequalto(1000)/(81)$.

8. True or False: To prepare a 50 ml solution of pH 1.25 HCl, one should add 49.76 ml of H_2O to 0.24 ml of concentrated HCl.

9. True or False: The molarity of the sodium hydroxide solution is 0.250 M, which is a common concentration for strong bases.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the pH of a solution formed by mixing 50.0 mL of 0.0025 M HBr with 50.0 mL of 0.0023 M KOH. Discuss the underlying principles and calculations involved in your answer.

12. A particle of mass 9.1×10^{-28} g in a one-dimensional box transitions from the $n = 5$ to $n = 2$ level, emitting a photon with frequency 6.0×10^{14} s^{-1}. Calculate the length of the box and discuss the implications of your findings in the context of quantum mechanics.

End of Paper